

2ÈME ANNÉE CYCLE D'INGÉNIEUR - SPÉCIALITÉ INFORMATIQUE.

MATIÈRE :
BASES DE DONNÉES AVANCÉES

Compte rendu TP2

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1 Exercice 1 : Requêtes

1.1 Question 1

Trouver le département avec le budget le plus élevé.

Listing 1 – Requête Q1

```
SELECT dept_name
FROM department
WHERE budget IN (
SELECT MAX(budget)
FROM department
);
```

Réponse : Finance

1.2 Question 2

Trouver les enseignants dont le salaire est supérieur à la moyenne des salaires de tous les enseignants.

Listing 2 – Requête Q2

```
SELECT Name_A.TEACHER_NAME, Name_A.salary
FROM TEACHER Name_A
WHERE Name_A.salary > ( SELECT AVG(B.salary) FROM TEACHER B);
```

Remarque : Le *FROM Teacher AS B* ne marche qu'en *PL/SQL* d'Oracle, mais pas dans *APEX Oracle*.

TEACHER_NAME	SALARY
Wu	90000
Einstein	95000
Gold	87000
Katz	75000
Singh	80000
Brandt	92000
Kim	80000

7 rows returned in 0.02 seconds [Download](#)

FIGURE 1 – Résultat de la requête Q2 (Salaires supérieurs à la moyenne)

1.3 Question 3

Trouver les paires (enseignant, étudiant) telles que l'enseignant ait donné au moins deux cours différents à l'étudiant.

Listing 3 – Requête Q3

```
SELECT teacher.teacher_name, student.student_name, COUNT()
FROM teacher, student, takes, teaches
WHERE teacher.teacher_id = teaches.teaches_id
AND student.student_id = takes.takes_id
```

```
AND takes.course_id = teaches.course_id
AND takes.sec_id = teaches.sec_id
AND takes.semester = teaches.semester
AND takes.takes_year = teaches.teaches_year
GROUP BY teacher.teacher_name, student.student_name
HAVING COUNT() >= 2;
```

TEACHER_NAME	STUDENT_NAME	COUNT(*)
Srinivasan	Bourikas	2
Srinivasan	Shankar	3
Crick	Tanaka	2
Katz	Levy	2
Srinivasan	Zhang	2

1.4 Question 4

Alternative pour Q3.

Listing 4 – Requête Q4

```
SELECT T.teachername, T.studentname, T.totalcount
FROM (
SELECT
teacher.teacher_name AS teachername,
student.student_name AS studentname,
COUNT(*) AS totalcount
FROM
teacher
JOIN teaches ON teacher.teacher_id = teaches.teaches_id
JOIN takes ON takes.course_id = teaches.course_id
AND takes.sec_id = teaches.sec_id
AND takes.semester = teaches.semester
AND takes.takes_year = teaches.teaches_year
JOIN student ON student.student_id = takes.takes_id
GROUP BY
teacher.teacher_name, student.student_name
) T
WHERE T.totalcount >= 2
ORDER BY T.teachername;
```

TEACHERNAME	STUDENTNAME	TOTALCOUNT
Crick	Tanaka	2
Katz	Levy	2
Srinivasan	Bourikas	2
Srinivasan	Shankar	3
Srinivasan	Zhang	2

1.5 Question 5

Trouver les étudiants qui n'ont suivi aucun cours avant 2009.

Listing 5 – Requête Q5

```
SELECT student.student_id, student.student_name
FROM student
MINUS
SELECT student.student_id, student.student_name
```

```
FROM student
JOIN takes ON takes.takes_id = student.student_id
WHERE takes.takes_year < 2009;
```

STUDENT_ID	STUDENT_NAME
00128	Zhang
12345	Shankar
19991	Brandt
23121	Chavez
44553	Peltier
45678	Levy
54321	Williams
55739	Sanchez
70557	Snow
76543	Brown
76653	Aoi
98765	Bourikas
98988	Tanaka

1.6 Question 6

Trouver les enseignants dont le nom commence par 'E'.

Listing 6 – Requête Q6

```
SELECT * FROM teacher WHERE teacher_name LIKE 'E%';
```

TEACHER_ID	TEACHER_NAME	DEPT_NAME	SALARY
22222	Einstein	Physics	95000
32343	El Said	History	60000

1.7 Question 7

Trouver le nom de l'enseignant ayant le 4ème plus haut salaire.

Listing 7 – Requête Q7

```
SELECT teacher_name
FROM teacher T1
WHERE 3 = (
SELECT COUNT(DISTINCT T2.salary)
FROM teacher T2
WHERE T2.salary > T1.salary
);
```

Réponse : Gold

1.8 Question 8

Trouver les noms et salaires des 3 enseignants ayant les plus bas salaires.

Listing 8 – Requête Q8

```
SELECT T1.teacher_name, T1.salary
FROM teacher T1
```

```
WHERE 2 >= (
SELECT COUNT(DISTINCT T2.salary)
FROM teacher T2
WHERE T2.salary < T1.salary
)
ORDER BY T1.salary ASC;
```

TEACHER_NAME	SALARY
Mozart	40000
El Said	60000
Califieri	62000

1.9 Question 9

Trouver les étudiants qui ont suivi un cours durant 'Fall' 2009 (en utilisant IN).

Listing 9 – Requête Q9

```
SELECT S.student_name
FROM student S
WHERE ('Fall', 2009) IN (
SELECT takes.semester, takes.takes_year
FROM takes
WHERE takes.takes_id = S.student_id
);
```

STUDENT_NAME
Zhang
Shankar
Peltier
Levy
Williams
Brown
Bourikas

1.10 Question 10

Trouver les étudiants qui ont suivi un cours durant 'Fall' 2009 (en utilisant EXISTS).

Listing 10 – Requête Q10

```
SELECT S.student_name
FROM student S
WHERE EXISTS (
SELECT 1
FROM takes
WHERE takes.takes_id = S.student_id
AND takes.semester = 'Fall'
AND takes.takes_year = 2009
);
```

STUDENT_NAME
Zhang
Shankar
Peltier
Levy
Williams
Brown
Bourikas

1.11 Question 11

Trouver les étudiants qui ont suivi un cours durant 'Fall' 2009 (en utilisant NATURAL JOIN).

Listing 11 – Requête Q11

```
SELECT DISTINCT student.student_name
FROM takes
NATURAL JOIN student
WHERE takes.semester = 'Fall'
AND takes.takes_year = 2009;
```

STUDENT_NAME
Brandt
Sanchez
Brown
Aoi
Zhang
Levy
Snow
Bourikas
Shankar
Chavez
Peltier
Tanaka
Williams

1.12 Question 12

Trouver les étudiants qui ont suivi un cours durant 'Fall' 2009 (en utilisant EXISTS - alternative).

Listing 12 – Requête Q12

```
SELECT student.student_name
FROM student
WHERE EXISTS (
SELECT 1
```

```
FROM takes
WHERE takes.takes_id = student.student_id
AND takes.semester = 'Fall'
AND takes.takes_year = 2009
);
```

STUDENT_NAME
Zhang
Shankar
Peltier
Levy
Williams
Brown
Bourikas

1.13 Question 13

Trouver les paires d'étudiants (nom1, nom2) qui ont suivi exactement les mêmes cours (même section, semestre, année).

Listing 13 – Requête Q13

```
SELECT Name_A.student_name, Name_B.student_name
FROM student Name_A
JOIN takes TA ON Name_A.student_id = TA.takes_id
JOIN student Name_B ON Name_B.student_id -- Correction: Should likely join based
    on shared courses, not student ID again
JOIN takes TB ON Name_B.student_id = TB.takes_id
WHERE Name_A.student_id <> Name_B.student_id
AND TA.course_id = TB.course_id
AND TA.sec_id = TB.sec_id
AND TA.semester = TB.semester
AND TA.takes_year = TB.takes_year
AND Name_A.student_name < Name_B.student_name
-- This query needs refinement to ensure exactly the same courses are taken, not
    just one common course.
-- The original query likely finds pairs sharing at least one section.
ORDER BY Name_A.student_name, Name_B.student_name;
```

Réponse : No Data Found (selon le document original)